The Contact Breaks at Horizon Five

Exact separation and an all-horizon logarithmic lower bound

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For the quartic pair

$$P = (\tfrac{1}{8}, \tfrac{1}{4}, \tfrac{1}{4}, \tfrac{1}{4}, \tfrac{1}{8}), \qquad Q = (\tfrac{7}{64}, \tfrac{5}{16}, \tfrac{5}{32}, \tfrac{5}{16}, \tfrac{7}{64}),$$

we have $G_P(z) - G_Q(z) = (1-z)^4/64$ and $\mathbb{E}P = \mathbb{E}Q = 2$. An exact depth-$n$ tree consists of probability laws B_h on $\mathbb{Z}_{\geq 0}$, with finite first moments, satisfying

$$Q * B_h = \sum_{x=0}^{4} P(x)\, \delta_x * B_{hx} \qquad (|h| < n). \tag{1}$$

Let $\mathcal{B}_r$ be the minimum root mean at depth r, with $\mathcal{B}_0 = 0$, and set $\Delta_h = \mathbb{E}B_h - \mathcal{B}_{n-|h|} \geq 0$. The earlier certified contacts at depths three and four remain valid. Their proposed extension to all larger horizons does not.

Theorem 1 (Exact failure at depth five)**.** *The depth-five optimum satisfies the certified rational bounds*

$$\boxed{0.3783154 < \mathcal{B}_5 < 0.3783157.} \tag{2}$$

Let $\mathcal{C}_5$ be the infimum root mean among exact depth-five trees with $\mathbb{E}B_{(4)} = \mathcal{B}_4$. Then

$$\boxed{\mathcal{C}_5 > 0.3812337, \qquad \mathcal{C}_5 - \mathcal{B}_5 > 0.002918.} \tag{3}$$

All terminating decimals in these inequalities denote exact rationals. In particular, every *optimal depth-five tree satisfies*

$$\boxed{\Delta_{(4)} > 0.} \tag{4}$$

Proof. First, the finite support reduction applies at every finite horizon. Equation (1) has a causal lift with fresh independent $Y \sim Q$: given $J_t = j$ and output history h, choose X using

$$K_h(x \mid j+y) = \frac{P(x) B_{hx}(j+y-x)}{(Q * B_h)(j+y)}$$

where the denominator is positive, and put $J_{t+1} = J_t + Y - X$. For a depth-n trajectory subtract $D = (J_0 - 4n)_+$ from every reserve while keeping the full X, Y trajectory unchanged. The shifted reserve is nonnegative, since $J_t \geq J_0 - 4t$. Fresh inputs and the unconditional iid output law are preserved. The root mean strictly decreases if it has mass above $4n$. A nonempty compact finite LP therefore attains the unrestricted minimum, and every optimizer has root support in $0, \ldots, 4n$.

Second, the previously certified value is

$$\mathcal{B}_4 = \frac{144958044813242692}{420381282815550045}.$$

In its supplied baseline dual, the reduced costs of root atoms $j = 0, 1$ are zero and those of every $j = 2, \ldots, 16$ are strictly positive. Complementary slackness and the root mean thus force the unique optimal root law

$$B_4^* = (1 - \mathcal{B}_4)\delta_0 + \mathcal{B}_4 \delta_1. \tag{5}$$

Consequently, depth-five contact at (4) is equivalent to prescribing $B_{(4)} = B_4^*$. Root truncation at 20 preserves this contact: the shifted child mean is $\mathcal{B}_4 - \mathbb{E}[D \mid X_1 = 4]$, while its feasible depth-four subtree requires a mean at least $\mathcal{B}_4$. Hence this conditional shift has mean zero.

Finally, the new exact certificates described on page 3 provide a feasible depth-five root mean $U < 0.3783157$, a baseline dual lower bound exceeding 0.3783154, and a dual lower bound $L > 0.3812337$ under (5). Thus $\mathcal{B}_5 \leq U < L \leq \mathcal{C}_5$, with $L - U > 1459/500000 = 0.002918$. This proves all three assertions. $\square$

An analytic lower bound for every horizon

The failure of the particular contact path does not invalidate the general accounting identity. More strongly, the same tree equations force unbounded reserve growth without assuming any contact.

Theorem 2 (Logarithmic reserve growth)**.** *Fix $0 < z < 1$ and write*

$$p = G_P(z), \qquad q = G_Q(z), \qquad c(z) = \frac{\log(p/q)}{-\log z} > 0.$$

For all integers $r \geq t \geq 1$,

$$\boxed{\mathcal{B}_r \geq (1 - p^t)\mathcal{B}_{r-t} + t\, c(z)\, p^t.} \tag{6}$$

Consequently, for every $n \geq 1$,

$$\boxed{\mathcal{B}_n \geq \max_{1 \leq t \leq n} t\, c(z) \left[1 - (1 - p^t)^{\lfloor n/t \rfloor}\right].} \tag{7}$$

In particular,

$$\boxed{\liminf_{n \to \infty} \frac{\mathcal{B}_n}{\log n} \geq \frac{\log(G_P(z)/G_Q(z))}{(-\log z)(-\log G_P(z))} > 0.} \tag{8}$$

All logarithms are natural.

Proof. Consider an optimal depth-r tree, and put $F_h(z) = \mathbb{E}[z^{J_h}]$, $m_h = \mathbb{E}B_h$, $b = \mathcal{B}_{r-t}$, and $a = -\log z > 0$. Iterating (1) through t generations gives

$$q^t F_\varnothing(z) = \sum_{|g|=t} P(g) z^{\sum_i g_i} F_g(z) = p^t \mathbb{E}_{P_z^{\otimes t}} F_g(z), \tag{9}$$

where $P_z(x) = P(x) z^x / p$. Jensen's inequality, first within each reserve law and then over the tilted histories, yields

$$\mathbb{E}_{P_z^{\otimes t}} F_g(z) \geq \mathbb{E}_{P_z^{\otimes t}} e^{-a m_g} \geq e^{-a \mathbb{E}_{P_z^{\otimes t}} m_g}.$$

Since $F_\varnothing(z) \leq 1$, equation (9) implies

$$\mathbb{E}_{P_z^{\otimes t}} m_g \geq \frac{t \log(p/q) - \log F_\varnothing(z)}{a} \geq t\, c(z). \tag{10}$$

Subtree feasibility gives $m_g - b \geq 0$. Moreover

$$\frac{P_z^{\otimes t}(g)}{P(g)} = \frac{z^{\sum_i g_i}}{p^t} \leq p^{-t}, \qquad \sum_{|g|=t} P(g) m_g = \mathcal{B}_r,$$

the last equality following by conservation of first moments. Therefore

$$\mathbb{E}_{P_z^{\otimes t}} m_g \leq b + p^{-t}(\mathcal{B}_r - b).$$

Combine this with (10) to obtain (6). Iterating at $r = t, 2t, \ldots, kt$ gives $\mathcal{B}_{kt} \geq t c(z)[1 - (1 - p^t)^k]$. Pruning a tree shows that $\mathcal{B}_n$ is nondecreasing, proving (7).

For the asymptotic claim, take

$$t_n = \left\lfloor \frac{\log n - 2 \log\log n}{-\log p} \right\rfloor.$$

For large n, this is positive, $t_n / \log n \to 1/(-\log p)$, and $\lfloor n/t_n \rfloor p^{t_n} \to \infty$. Hence $(1 - p^{t_n})^{\lfloor n/t_n \rfloor} \to 0$ in (7), which proves (8). $\square$

For the explicit rational tilt $z = 1/16$,

$$p = \frac{74273}{524288}, \quad q = \frac{543559}{4194304}, \qquad \liminf_{n \to \infty} \frac{\mathcal{B}_n}{\log n} \geq \underbrace{\frac{\log(594184/543559)}{\log 16\, \log(524288/74273)}}_{\kappa_0 \approx 0.0164347253}.$$

Thus $\mathcal{B}_n \to \infty$. In particular, no exact infinite tree in this model can have finite root mean: its every finite truncation would require at least $\mathcal{B}_n$ reserve.

The exact depth-five certificate system

Use conditional-law variables $u_{h,j} = B_h(j)$, with histories in increasing length and lexicographic order and

$$0 \le j \le M_h := 20 + 4|h| - \sum_i h_i.$$

This upper bound follows from $J_t = J_0 + \sum_{i \le t} Y_i - \sum_{i \le t} X_i$. The full LP has $u \ge 0$, one normalization $\sum_j u_{h,j} = 1$ per history, and, for each internal history and each $0 \le s \le M_h + 4$,

$$\sum_{y=0}^{4} (64Q(y)) u_{h,s-y} - \sum_{x=0}^{4} (64P(x)) u_{hx,s-x} = 0,$$

with out-of-range variables equal to zero. Its objective is $\min \sum_j j u_{\varnothing,j}$. Rows consist of all normalizations first, followed by convolution rows in history order and increasing s.

For the contact LP, remove histories beginning with 4 and substitute (5) in the root equation. Its only nonzero convolution right-hand sides are $8(1 - \mathcal{B}_4)$ at $s = 4$ and $8\mathcal{B}_4$ at $s = 5$. An optimal depth-four subtree with the prescribed root can always be attached, so this reduced LP describes exactly the contact constraint.

LP	Variables	Rows	Zero dual rc	Positive dual rc
Full depth five	119,136	29,291	56,406	62,730
Depth five with (4) contact	96,875	23,750	45,013	51,862

The new certificates give the exact feasible upper bound

$$U = \frac{1034773180444807023527498014315484839916547139253}{273521084485670813228916005827574062592234909610 0},$$

and the exact dual lower bounds

$$L_0 = \frac{53243169418655}{140737488355328}, \qquad L = \frac{451101764942215223297512112410 09}{118326811790102642263741452779520}.$$

Every primal normalization and convolution equation holds exactly. Both duals satisfy $c - A^T y \ge 0$ at every column and have objective $b^T y = L_0$ or L. Their reduced-cost counts are shown above; neither has a negative reduced cost. Weak duality proves $L_0 \le \mathcal{B}_5 \le U$ and $L \le \mathcal{C}_5$. An exact value of $\mathcal{B}_5$ is not needed for the separation theorem.

Reproduction and the remaining asymptotic question

The accompanying package contains the original four-horizon certificates, the three new rational JSON witnesses, the independent verification scripts, their successful logs, and the paper sources. From the extracted package root, run

```
python verify_all.py
```

Verification requires only Python's standard library and exact `fractions.Fraction` arithmetic. No optimizer or numerical tolerance enters verification. Floating-point optimization was used to discover the new witnesses; the stored rational witnesses are checked separately.

There is also an elementary upper bound. Concatenate a depth-m tree A and a depth-n tree C by using $A_h * C_\varnothing$ in the first block and $A_h * C_g$ after a length-m history h. Convolution preserves (1), and the root means add. Thus

$$\mathcal{B}_{m+n} \le \mathcal{B}_m + \mathcal{B}_n, \qquad \mathcal{B}_n \le \lfloor n/5 \rfloor U + \mathcal{B}_{n \bmod 5}.$$

Writing $n = ak + r$ proves directly that

$$\lim_{n\to\infty} \frac{\mathcal{B}_n}{n} = \inf_{k \ge 1} \frac{\mathcal{B}_k}{k} =: \beta, \qquad 0 \le \beta \le U/5 \approx 0.0756631382.$$

The established all-horizon bounds are therefore $\mathcal{B}_n = \Omega(\log n)$ and $\mathcal{B}_n = O(n)$. They do not determine β, prove $\Theta(\log n)$, or give a sharp asymptotic equivalent. The specific universal (4)-contact claim is disproved; the sharp growth rate remains unresolved here.